Predicting NEOWISE-Derived Diameters of Solar System Small Bodies Using Physics-Informed Machine Learning

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Abstract: Accurate diameter estimation is essential for characterizing small bodies throughout the Solar System, with particular importance for assessing the potential impact hazards posed by near-Earth objects (NEOs). However, uncertainties in observational properties such as absolute magnitude and geometric albedo complicate the determination of Solar System small body sizes. This study investigates whether physics-informed machine learning can improve predictions of NEOWISE-derived small body diameters compared with a conventional physics-based estimator. Using 181,965 observations from the NEOWISE catalog, Random Forest and XGBoost regression models were evaluated with two feature configurations: one using absolute magnitude and geometric albedo, and another incorporating seven additional features, including a physics-derived diameter estimate. Model performance was assessed using mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). While the physics-based estimator outperformed the two-feature machine-learning models in test MAE, the hyperparameter-tuned nine-feature Random Forest achieved the best overall test performance, reducing MAE by 2.6% (from 0.152 to 0.148 km) and RMSE by 16.9% (from 1.198 to 0.995 km) relative to the physics-based estimator, while increasing R^2 from 0.990 to 0.993. Additional analyses revealed that Random Forest generally benefited from increased training data, while prediction errors were more pronounced at lower absolute magnitudes and geometric albedos. SHAP analysis further identified the physics-derived diameter estimate as the most influential predictor in the nine-feature model. These findings suggest that while conventional physical relationships remain strong predictors of diameter, incorporating them into machine-learning models alongside additional observational features can further reduce prediction errors, particularly larger deviations from catalog-derived estimates. Rather than replacing established physical models, this research highlights the potential of physics-informed machine learning to refine small body characterization using existing observational data.

Keywords: Solar System Small Bodies, Diameter Estimation, NEOWISE, Machine Learning, Physics-Informed Machine Learning, Random Forest, XGBoost, Geometric Albedo, Absolute Magnitude, SHAP Analysis

1. Introduction

Solar System small bodies, a collection of populations that orbit the Sun, have been a large topic of scientific development within the past several decades. Near-Earth objects (NEOs), specifically, are classified as asteroids or comets whose perihelion distance is 1.3 AU or less. Near-Earth asteroids make up a majority of this population, while the remaining 5-15% of the population belongs to comets (1). Both smaller and larger NEOs can cause significant destruction, which has driven the development of impact monitoring and detection systems (2) and of hazard-assessment tools such as the Palermo Technical Impact Hazard Scale (3). Despite the evolution of these planetary-defense algorithms, estimating the true size of a Solar System small body remains a persistent source of uncertainty as it is primarily calculated via ground-based estimation of albedo, the amount of light that a body reflects at a zero phase angle, and absolute magnitude, the measure of the true brightness of a small body.

Infrared-based surveys such as the NEOWISE mission seek to provide more accurate thermally modeled estimations of small bodies directly from space, but only for a small known population of data. Machine learning offers methods to train and develop algorithms on the relationships within the catalog-derived estimates. This study seeks to address the question of how accurately machine-learning models can reproduce NEOWISE-derived small body diameter estimates using cataloged physical and observational features, and whether incorporating a physics-based estimate improves model performance? For this study's purposes, I utilized the NEOWISE Derived Diameters and Albedos of Solar System Small Bodies Catalog Definitions V2 dataset (4) made available by NASA/IPAC. The complete analysis code and computational notebook will be made publicly available upon acceptance of the manuscript.

2. Materials and Methods

2.1 Dataset

This NEOWISE dataset used in this study contains physical and observational measurements derived from the Wide-field Infrared Survey Explorer telescope (5), where NEOWISE was the reactivation of the WISE telescope to focus on small bodies within the Solar System. The dataset consists of several Solar System small body populations and is predominantly composed of main-belt asteroids. NEOWISE measurements have been used extensively in previous studies of small body physical properties, including diameter and albedo estimation (6,1)

The original dataset contained 183,412 observations. Prior to model development, observations with missing values for absolute magnitude (H), geometric albedo (pV), and slope parameter (G) were removed, leaving 181,965 observations for analysis. These variables were required for the principal model configurations and for calculation of the physics-based diameter estimate.

The processed dataset was randomly divided into training, validation, and test sets using an 80%/10%/10% split with a fixed random seed of 42. This resulted in 145,572 training

observations, 18,196 validation observations, and 18,197 test observations. The training set was used for model fitting and cross-validated hyperparameter optimization, while the validation and test sets were retained for out-of-sample performance assessment.

2.2 Baseline Methods

Three baseline approaches were evaluated for small body diameter prediction: a physics-based estimator, Random Forest regression, and XGBoost regression. The physics-based model provides a conventional reference derived from the established relationship between absolute magnitude, geometric albedo, and diameter (7). Because geometric albedo and diameter are physically and computationally related in the NEOWISE modeling framework, using cataloged albedo as an input introduces dependence between the predictors and the reference diameter. Consequently, the results should be interpreted as reproducing catalog-derived diameter estimates rather than independently measuring sizes. Random Forest and XGBoost were included as nonlinear tree-based machine-learning methods capable of modeling interactions among multiple small body characteristics (8,9).

The physics-based baseline estimated small body diameter using

$$D = \frac{1329}{\sqrt{pV}} \times 10^{-0.2H} \quad \text{(eq1)}$$

where D is diameter in kilometers, H is absolute magnitude, and pV is geometric albedo. This equation directly estimates diameter from two measured physical properties and does not require model training. It therefore served as a physically motivated reference against which the machine-learning models were compared.

Random Forest regression constructs an ensemble of decision trees using randomized subsets of the training data and features. Predictions from the individual trees are combined to obtain the final regression estimate. This ensemble structure can model nonlinear relationships and interactions without requiring an explicit functional relationship between the input variables and small body diameter. Both a default Random Forest model and a hyperparameter-tuned variant were evaluated.

XGBoost is a gradient-boosted decision-tree method in which trees are added sequentially, with each new tree trained to reduce errors made by the preceding ensemble. This approach allows the model to learn complex nonlinear relationships while controlling model complexity through parameters such as tree depth, learning rate, and subsampling. As with Random Forest, both default and hyperparameter-tuned XGBoost models were evaluated.

Importantly, two feature configurations were used to assess whether additional observational information improved diameter prediction. The two-feature configuration contained only absolute magnitude (H) and geometric albedo (pV). These variables were selected as they are the same quantities used by the physics-based diameter equation, allowing a direct comparison between the analytical relationship and machine-learning models trained on equivalent information.

The nine-feature configuration expanded the model inputs to include absolute magnitude, geometric albedo, slope parameter, the numbers of observations in the four WISE wavelength bands, geometric-albedo uncertainty, and the diameter predicted by the physics-based equation. Including the diameter predicted by the physics-based equation provided the machine-learning models with an explicitly physics-derived estimate while allowing them to learn corrections or additional relationships from the remaining observational variables. The two-feature and nine-feature configurations were evaluated using the same Random Forest and XGBoost modeling framework so that differences in performance could be attributed to the information available to the models rather than to changes in model type.

2.3 Metrics

Model performance was evaluated using mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). These metrics are commonly used together in regression studies because they provide complementary measures of predictive accuracy and model fit. MAE has been used in prior regression-based scientific studies to quantify average prediction error (10). RMSE is also commonly reported because it gives greater weight to larger prediction errors (11). R^2 is frequently used to describe how much of the variation in observed values is explained by a regression model (12).

Mean absolute error (MAE) was calculated as:

$$MAE = \frac{1}{n}\sum\left|y_i - \hat{y}_i\right| \quad \text{(eq2)}$$

where y_i is the observed diameter, $\hat{y}_i$ is the predicted diameter, and n is the number of observations. MAE represents the average magnitude of the prediction error in the original units of the target variable, kilometers. Lower MAE values indicate more accurate predictions.

Root mean squared error (RMSE) was calculated as:

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2} \quad \text{(eq3)}$$

Because prediction errors are squared before averaging, RMSE places greater emphasis on larger prediction errors than MAE. RMSE is also expressed in kilometers, allowing it to be interpreted directly in terms of small body diameter prediction error. Lower RMSE values indicate better predictive performance.

The coefficient of determination (R^2) was calculated as:

$$R^2 = 1 - \frac{\Sigma(y_i - \hat{y})^2}{\Sigma(y_i - \bar{y})^2} \quad \text{(eq4)}$$

where $\bar{y}$ is the mean observed small body diameter. R^2 measures the proportion of variation in observed diameters explained by the model, with values closer to 1 indicating a stronger fit.

MAE, RMSE, and R^2 were reported together to evaluate model performance from complementary perspectives: average prediction error, sensitivity to larger errors, and overall explained variance.

2.4 Implementation Details

All experiments were implemented in Python and executed in Google Colab with an NVIDIA T4 GPU. A fixed random seed of 42 was used for data splitting, model initialization, randomized hyperparameter search, and random subsampling to ensure reproducibility. Random Forest and XGBoost models were evaluated using both default and tuned configurations. Hyperparameters were optimized on the training set using RandomizedSearchCV with 25 sampled configurations, three-fold cross-validation, and negative mean squared error as the selection criterion. For Random Forest, the search considered 100-300 trees, maximum depths of 5-15, minimum leaf sizes of 1-5, minimum split sizes of 2-10, and feature-sampling fractions of 0.6-1.0. For XGBoost, the search considered maximum depths of 3-6, learning rates of 0.05-0.1, 100-300 estimators, subsampling fractions of 0.8-1.0, minimum child weights of 1-3, and column-sampling fractions of 0.8-1.0. The held-out test set was not used for hyperparameter selection.

3. Results

3.1 Baselines

When analyzing the baseline methods listed in Table 1, the nine input-feature Random Forest model with hyperparameter tuning is highlighted as having the least error when performing against the testing data, even outperforming the physics-based method. The most notable improvement over the physics-based method is the Root Mean Square Error, with 0.995 km compared to 1.198 km from the physics formula. Because RMSE places greater weight on larger errors, this indicates that the Random Forest primarily reduced the largest errors. The improvement in Mean Absolute Error was smaller between the two methods, with 0.148 km

compared to the physics-based method's 0.152 km. Furthermore, hyperparameter tuning produced only a small gain over the default nine input-feature Random Forest on the testing data (MAE of 0.148 km versus 0.149 km, RMSE of 0.995 km versus 1.039 km). The default Random Forest model incurs a 176% increase in MAE from training to testing for the nine input-feature model, versus only a 78% MAE increase for the same model with two input-features, which is indicative of a training-to-test generalization gap with a larger number of input-features that may be consistent with overfitting. However, the Random Forest tuned model comparatively derived a 32% and 27% increase in MAE between nine input-features and two input-features, respectively, demonstrating a far smaller training-to-test generalization gap than the default Random Forest model. Additionally, both XGBoost methods, in both input-feature configurations, have much greater error against the testing data than all of the other methods. Random Forest trains deep independent decision trees in parallel, averaging the results to stay within the range of the data, while XGBoost builds shallow sequential trees that fit based on residual error. Differences in ensemble structure, parameter settings, and sensitivity to the training data may have contributed to the larger error results of XGBoost, although the present experiments do not isolate their individual effects. Among the two-feature models, the physics-based formula achieved the lowest test MAE, despite the lower training errors of the Random Forest models. This difference further suggests that the machine-learning models exhibited larger training-to-test generalization gaps.

Table 1. Baseline Method error results, where (+) indicates hyperparameter-tuned models. Best results are **bolded**, and second-best results are *italicized*.

Number of Input Features	Methods	Train			Val			Test		
		MAE	*RMSE*	*R²*	*MAE*	*RMSE*	*R²*	*MAE*	*RMSE*	*R²*
2	Small Body Size Estimator Formula	0.154	1.277	0.989	0.172	1.495	0.986	0.152	1.198	0.990
	Random Forest	*0.105*	*0.552*	*0.998*	0.194	1.303	0.989	0.187	1.136	0.991
	Random Forest (+)	0.140	0.892	0.995	0.190	1.494	0.986	0.178	1.079	0.992
	XGBoost	0.285	2.453	*0.959*	0.347	3.480	0.923	0.347	3.254	0.930
	XGBoost (+)	0.345	2.733	0.949	0.370	3.313	0.930	0.365	3.032	0.939
9	Small Body Size Estimator Formula	0.154	1.277	0.989	0.172	1.495	0.986	0.152	1.198	0.990
	Random Forest	**0.054**	**0.403**	**0.999**	**0.158**	**1.238**	**0.990**	*0.149*	*1.039*	*0.993*
	Random Forest (+)	0.112	0.619	0.997	*0.159*	*1.271*	*0.990*	**0.148**	**0.995**	**0.993**
	XGBoost	0.145	0.765	0.996	0.283	3.304	0.930	0.268	2.312	0.964
	XGBoost (+)	0.256	1.973	0.973	0.304	3.237	0.933	0.284	2.329	0.964

3.2 Dataset Size Ablation

When conducting dataset ablation across these methods, further insight is gained into the comparison between the Random Forest and XGBoost baselines, specifically. As seen in Figure 1, XGBoost tends to fluctuate in its error and precedence between both the default and hyperparameter models, not demonstrating a consistent pattern throughout the varying dataset sizes or number of input-features. Across increasing training-set sizes, the Random Forest models generally showed reductions in MAE and RMSE, although these improvements were not monotonic at every training fraction. Gains became smaller at higher training fractions, suggesting diminishing returns from additional observations within the evaluated range.

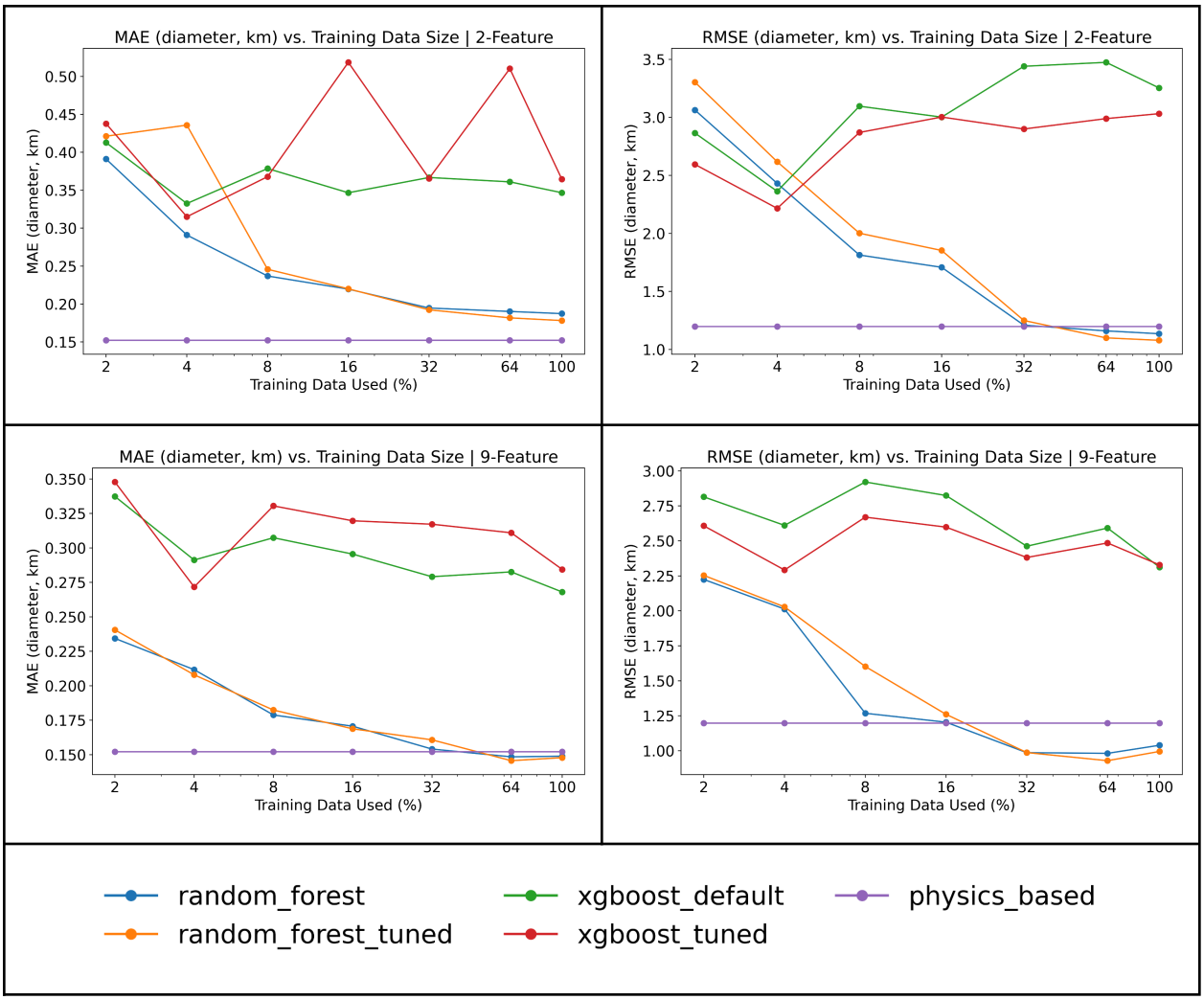


Figure 1. Testing data MAE and RMSE vs. training data sizes across two-feature and nine-feature baselines. Note: In Log scale.

3.3 Error Analysis

The signed residual distributions of diameter for the best-performing methods in Figure 2 indicate that larger residual magnitudes are concentrated among objects with lower absolute magnitudes and lower geometric albedos, while residual dispersion generally decreases as these feature values increase. For absolute magnitude, the varied residuals are prevalent around the values of 10-15, while the geometric albedo residuals are at around 0-0.1. It can also be noted that the concentration of residual variation is more spread out among the absolute magnitude distribution, and that both input-features' distributions funnel toward 0 as the value of the respective feature increases.

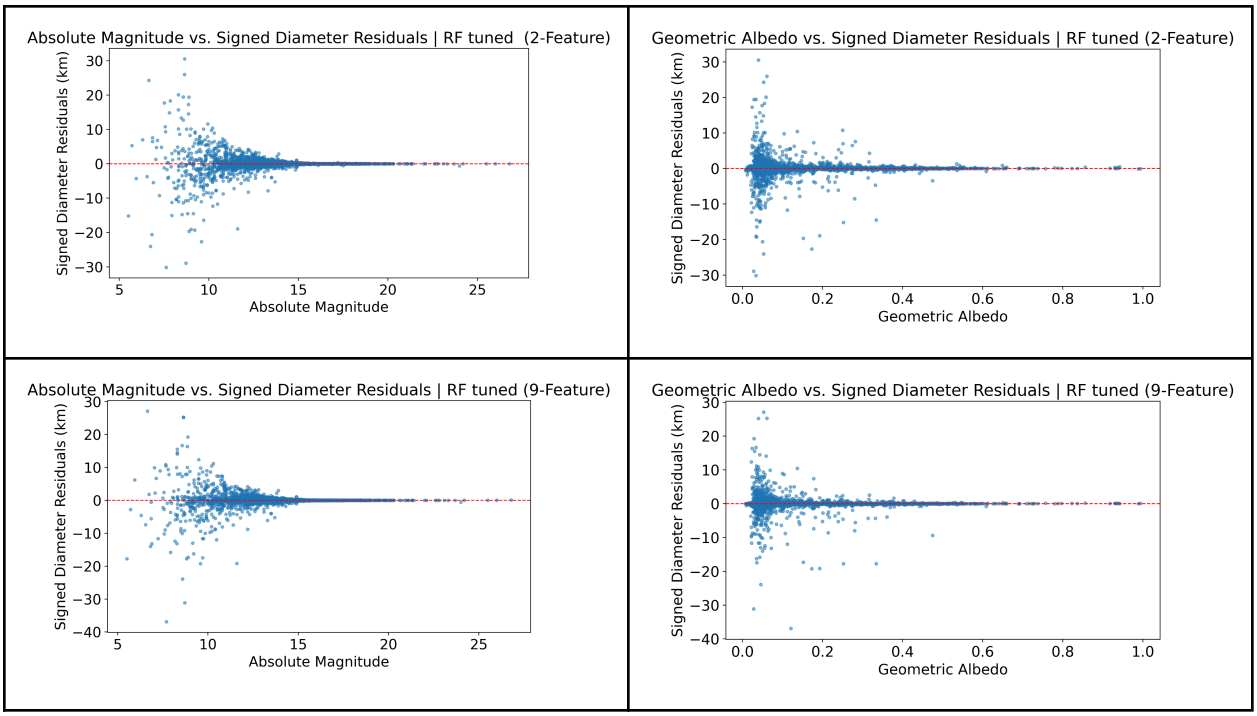


Figure 2. Signed diameter residuals (predicted - observed) for absolute magnitude and geometric albedo across the best-performing two-feature and nine-feature baselines.

Within a SHAP analysis of the best performing baseline in Figure 3, the physics-derived diameter feature (the output of the small body size estimation formula) contributed more to the model's predictions than any other input-feature. It should be noted that the SHAP calculations used a 2000-row random sample of the larger dataset in order to obtain this plot. Because the physics-derived diameter feature is a deterministic function of absolute magnitude and geometric albedo, SHAP importance values for these correlated inputs should be interpreted cautiously and not as independent contributions. Since absolute magnitude and geometric albedo are already a part of the physics-based formula, their input likely had a larger effect within the formula than when contributing individually to the model. Furthermore, among the WISE band observation counts, the number of W1 observations has higher feature importance than the others. Because this feature counts observations and does not measure infrared brightness, its importance may reflect how thoroughly an object was observed more than size information in the band itself. In addition to the physics-based estimator, the other input-features seem to serve a supporting role to the formula overall. Lastly, slope parameter had the least input towards the final predictions, possibly due to the fact that a large portion of the rows are the assumed value of 0.15 rather than an actually calculated slope parameter.

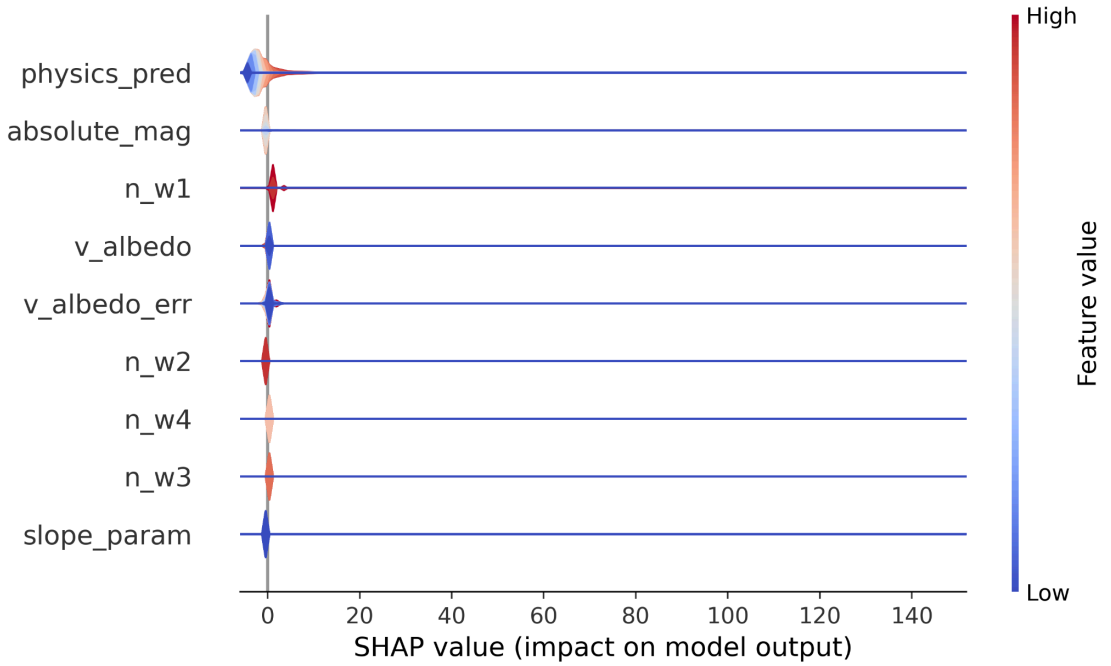


Figure 3. SHAP layered violin plot for feature importance of Random Forest tuned model with nine input-features.

4. Discussion

In impact consequence assessment, the impact energy of a near-Earth object scales approximately with the diameter to the third power when density and impact velocity are held constant, where larger NEOs pose a higher destructive risk but are more infrequent than smaller NEOs of <1 km in diameter (1). However, it should be noted that this study does not evaluate actual impact probabilities or planetary-defense performance of near-Earth objects. Across both input-feature configurations, Random Forest models were the strongest performing ML methods. The Random Forest tuned method with nine features, in particular, scored the lowest error in the testing set of data, with a MAE of 0.148 km, a RMSE of 0.995 km, and an R^2 of 0.993. This model also achieved lower errors on the testing set than that of the physics-based formula with a 0.004 km lower MAE, a 0.203 km lower RMSE, and a 0.003 higher R^2. This pattern indicates that Random Forest was able to reduce the largest error, while XGBoost performed as clearly less accurate than the other methods across both input-feature variations. A general training-to-test generalization gap was also apparent across the error values for all machine learning baselines, which may be consistent with overfitting. Additionally, as the training data was increased during ablation studies, Random Forest demonstrated the trend of improving its performance with a higher percentage of data, while XGBoost warranted no consistent trend. Signed residual errors of the diameters indicated that regardless of the method or feature, residual error was scattered towards lower values of H and pV, which correlate with larger small bodies, and concentrated towards 0 increasingly as the features increased in value. The SHAP calculations indicated that the physics-based feature was of high importance to the tuned nine-feature Random Forest model, using much of the absolute magnitude and albedo within the

formula (7). The other features take less importance, with absolute magnitude, W1 infrared band counts, and albedo being the most prominent factors in the predictions according to the SHAP plot. It's interesting that the W1 band count feature takes precedence, given that it records the number of observations of an object in that band and not its brightness. In NEOWISE thermal-model fits, W1 and W2 contain a mix of reflected sunlight and thermal emission for small bodies (6), but the importance of this count may more plausibly reflect how thoroughly an object was observed. The slope parameter had extremely little input towards the prediction, which could be a result of the data consisting primarily of an assumed value of 0.15 (13). The bagging vs boosting method difference between Random Forest and XGBoost, respectively, is a possible cause of the disparity in error between the two. With XGBoost's sequential tree building based on residual error of the previous trees, the results may be sensitive to hyperparameter settings. Noisy data may also be a factor for the higher error values within the XGBoost methods' results.

It should be acknowledged that this study predicts catalog-derived diameters, not independently verified physical diameters. The NEOWISE diameters are thermal-model fits based on the telescope's observations, where a spherical shape is assumed for small bodies (6). The catalog compiles several results from several analyses through NEOWISE with some values assumed rather than fitted (4). In addition, because catalog albedos are computed from the fitted diameter and absolute magnitude, using albedo as an input may introduce target-related information into the models. The models, therefore, may partly learn the catalog's internal relationship, which could contribute to the small differences between the learned models and the physics-based method.

The uncertainty of absolute magnitude and albedo measures still remains a limitation for discovery of brand-new small bodies, as this study was trained with measured values for both features as recorded by the NEOWISE mission. Because observations were randomly partitioned rather than explicitly grouped by small body identity, multiple observations of the same object may have occurred across subsets. Consequently, the reported performance may have overestimated generalization to previously unseen objects. Another shortcoming faced during the study is the loss of 1,447 rows of data that shared NaN values for absolute magnitude, geometric albedo, and slope parameter. Since the absolute magnitude and albedo values could not be properly replaced with an assumption, and given that the NaN rows were for all three of these features, I dropped the rows entirely from the dataset. It is possible that these values were NaN since their specific small bodies did not receive an adequate amount of light observation (6), although the cause of the missing values was not verified in this study. If so, this may lead the study's baselines to favor highly-characterized small bodies. Further, additional regularization methods were not systematically studied to augment the models, with depth restrictions, minimum leaf sizes, subsampling, and XGBoost's existing regularization mechanisms already constraining model complexity.

Future works should further pursue reducing the amount of error of larger diameter predictions and the training-to-test generalization gap found within this study's results, and further explore the domain of less characterized small bodies and how those values are handled within a larger study. Greater investigation of the XGBoost methods' hyperparameters may also yield different and potentially improved overall results. Additionally, model development on reducing the uncertainty of absolute magnitude and albedo measurements could provide insight on the still lingering question for the majority of small body observation studies.

5. Conclusion

This study investigated whether physics-informed machine learning could improve predictions of NEOWISE-derived diameters of Solar System small bodies. While the conventional physics-based estimator outperformed the two-feature machine-learning models in test MAE, the hyperparameter-tuned nine-feature Random Forest achieved the best overall performance, reducing MAE by 2.6% and RMSE by 16.9% relative to the physics-based estimator. Dataset size ablation showed that Random Forest generally benefited from additional training data, while SHAP analysis identified the physics-derived diameter estimate as the most influential predictor. These findings suggest that incorporating established physical relationships alongside additional observational features can improve machine-learning predictions, particularly by reducing larger deviations from catalog-derived diameters.

However, since NEOWISE-derived diameters and geometric albedo are closely related, improved agreement with catalog estimates does not necessarily indicate greater accuracy in determining true physical diameters. Future research should investigate performance using independently validated diameter measurements and less-characterized bodies. Overall, this study highlights the potential of physics-informed machine learning to complement conventional estimation methods and support further advances in small body characterizations.

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